Alan Lyu

Peking University junfenglyu7@gmail.com junfenglyu.github.io

Summary

I am Alan Lyu, an undergraduate student majoring in Physics at Peking University while also pursuing a second degree in Life Sciences. My research interests are biophysics and computational neuroscience. I use tools and frameworks from physics to study how complex structures and functions emerge in living systems. I hope to pursue doctoral study in biophysics, physics, or computational neuroscience.

Education

Peking University, Bachelor of Science *Sep. 2023 – Jul. 2027 (expected)*

- **Major:** Physics
- **Second Degree:** Life Science
- **CGPA:** 3.97/4.00*

> ** Note: Beginning in fall 2025, the School of Physics at Peking University no longer provides official GPA or cohort ranking. This GPA was calculated from the original grades of individual courses using the WES conversion method.*

- **Related Coursework:** 3.99 in Thermal Physics; 3.91 in Equilibrium Statistical Physics; 3.95 in Methods for Mathematical Physics; 3.97 in Electrodynamics; 3.93 in Computational Physics; 3.88 in Topics on Nonlinear Physics; A- in Physical Chemistry in Cellular and Molecular Biology; 3.95 in Lectures on Systems Biology; A in Fundamentals of Network Science and Systems Biology; A in Mathematical Modeling in the Life Sciences.

Standardized Tests

TOEFL iBT: 104/120

An updated TOEFL score will be available soon.

Research

Emergence and Learning Dynamics of Low-Rank Structure in Neural Networks

Advisors: Prof. Yuhai Tu and Dr. Yikuan Zhang

- Trained MLPs, CNNs, and Wilson–Cowan recurrent neural networks with biologically motivated dynamics at different sizes on MNIST, and characterized the emergence of low-rank structure in weight space across architectures and scales.
- Used singular value decomposition and related statistical analyses, together with nuclear-norm regularization and Hebbian plasticity, to investigate the learning dynamics underlying the emergence of low-rank structure.

Task-Driven Optimization of the *Drosophila* Olfactory Circuit for Joint Odor-Motion and Gradient Sensing

Advisors: Prof. Thierry Emonet and Dr. Kevin Chen

- Extended a recently reported model of odor-motion sensing into a parameterized circuit whose operating point and response properties can be optimized.
- Formulated joint odor-motion and odor-gradient sensing objectives under a metabolic constraint and optimized circuit parameters for performance on both tasks.
- Characterized how a shared network balances the two sensory objectives and identified parameter regimes that support robust encoding across changing odor environments.

Information-Theoretic Analysis of Environmentally Regulated Sessile–Motile State Switching in *Bacillus subtilis*

Advisor: Prof. Chao Tang

- Developed an agent-based model parameterized by experimentally measured switching rates to simulate sessile–motile state transitions in dynamic chemical environments.
- Applied transfer entropy and related information-theoretic measures to compare bidirectionally regulated agents with agents in which only the motile-to-sessile transition responds to the environment.
- Compared information acquisition and chemotactic performance to examine the adaptive value of alternative switching strategies in complex environments.

Teaching

Teaching Assistant (Dry Lab), Cell Biology Laboratory *Fall 2026*

School of Life Sciences, Peking University

- Guide undergraduate students in applying computer vision algorithms to identify stages of the cell cycle in HeLa cells labeled by immunofluorescence and in root tip cells of *Vicia faba* stained by the Feulgen method.

Honors and Awards

- National Scholarship of China & Peking University Merit Student Award 2024–2025
- National Scholarship of China & Peking University Merit Student Award 2023–2024
- Finalist in the Mathematical Contest in Modeling (MCM) 2026
- Meritorious in the Mathematical Contest in Modeling (MCM) 2025
- Peking University Undergraduate Physicists' Tournament (PKUPT) 2023–2024
  - Second Prize (Team rank 2/18)
  - Best Opponent and Best Reviewer Awards